CENG 424
Embedded Computer Systems
Final Presentation

PARTY BOT

G28

Spring 2026

Izmir Institute of Technology

Outline

- Meet the team
- Overview of the Project
- Specifications(H/s, S/w, tools, apps, etc.)
- Evaluation
- Business Opportunities
- Conclusion
- Demonstration

Team

- Timuçin TOPCU
  - Week 1-5-6-9-10 Tasks
- Emre ERTÜRK
  - Week 1-3-7-8 Tasks
- Onur KANSOY
  - Week 2-4-5-6-8 Tasks
- Tarık YARAŞLI
  - Week 2-3-5-6-9-10 Tasks









Overview of the Project

Introduction

The project involved designing a drivable indoor prototype controlled by kinematic data and discrete inputs processed from the user's hand movements. A robust, low-latency wireless communication protocol was established to ensure that the robot responds naturally to human input. The system uses a modular design consisting of a wearable sender-side glove and a receiver-side robotic platform. Both sides are built around the ESP32 DevKit V1, chosen for its dual-core processing power and integrated ESP-NOW protocol, which enables low-latency peer-to-peer wireless communication without requiring an external network.

Overview of the Project

Problem Definition

Current physical entertainment props present several limitations in terms of user interaction and flexibility. Many of these systems depend on fixed positions and simple activation methods, such as pressing a button or switching a device on and off. As a result, the interaction is usually short, predictable, and limited to a single action. This reduces sustained engagement because the user does not continuously control or interact with the system after activation.

Overview of the Project

Problem Definition

Another limitation is the lack of movement-based input. Traditional entertainment devices generally do not use natural hand gestures or body movements as part of the control process. This makes the interaction less dynamic and less engaging for the user. In addition, many props remain in one location and cannot easily adapt to moving crowds or changing indoor environments.

Overview of the Project

Solutions

PartyBot addresses these limitations by replacing static interaction with a gesture-controlled mobile platform. The system allows the user to continuously control the robot through hand movements using a wearable glove. It also provides additional interaction through music playback, mecanum wheel movement, strafe control, and visual feedback from the dot matrix display. Therefore, PartyBot offers a more dynamic and interactive entertainment experience compared to traditional fixed props.

Overview of the Project

Specifications

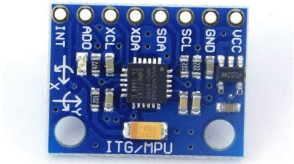
- Tools
- Hardware
- Software
- Applications
- Services

Picture of H/w and tools

Logos of s/w and apps

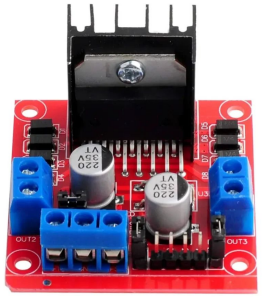
Hardware



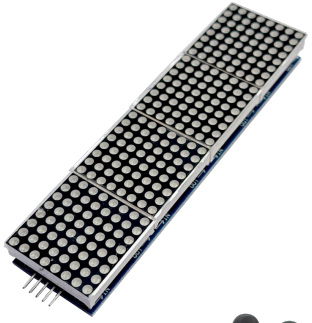

INT
ADO
XCL
XDA
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GND
VCC
ITG/MPU



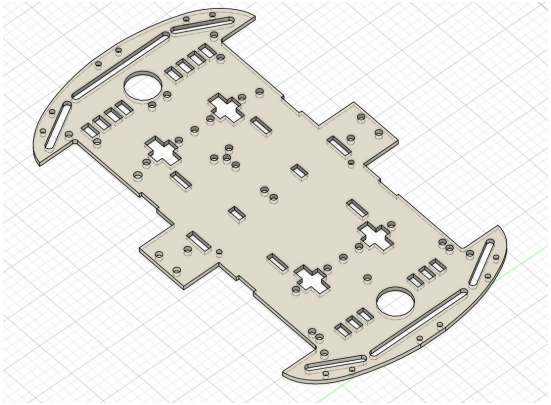



OUT2
OUT3





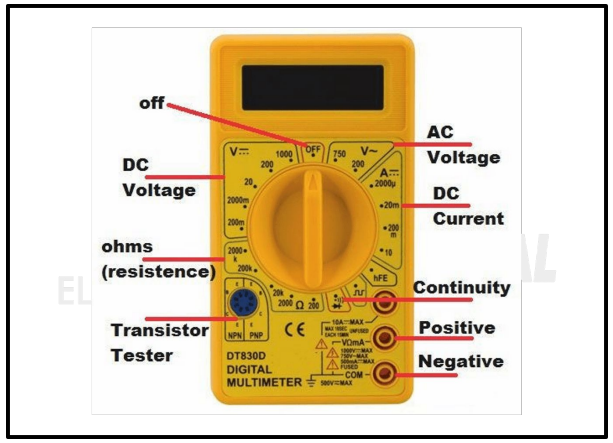

ORION
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Beybi
PN3 Zebra

Tools&Software







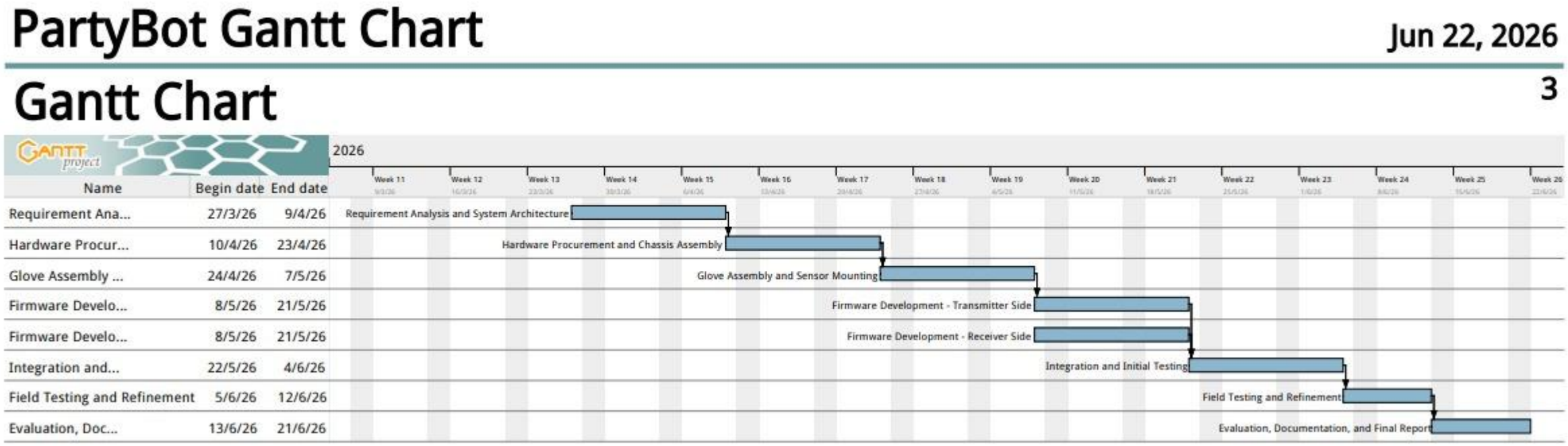




Arduino IDE

Evaluation

Gantt Chart



Evaluation
Timeline

Stage 1 - Requirement Analysis and System Architecture

Project requirements will be defined, overarching system architecture will be established, and basic communication protocols and command logic will be specified.

Stage 2 - Hardware Procurement and Chassis Assembly

Necessary components will be acquired. Mobile robotic platform will be assembled while integrating the drive systems, power supplies, and audio mechanism.

Stage 3 - Glove Assembly and Sensor Mounting

The wearable interface will be constructed by securely mounting and wiring all required kinematic sensors and push buttons to the glove structure.

Stage 4 - Firmware Development - Transmitter Side

Wearable controller will be programmed and will be ensured to read and filter sensor data, interpret user inputs, and transmit these as formatted data packets.

Stage 5 - Firmware Development - Receiver Side

Mobile base will be programmed to reliably receive commands, translate them into motor movements, and trigger the audio system. Basic safety fail-safes will be added.

Stage 6 - Integration and Initial Testing

Subsystems will be combined. Preliminary bench tests will be performed to verify end-to-end data transmission, motor control, and audio playback.

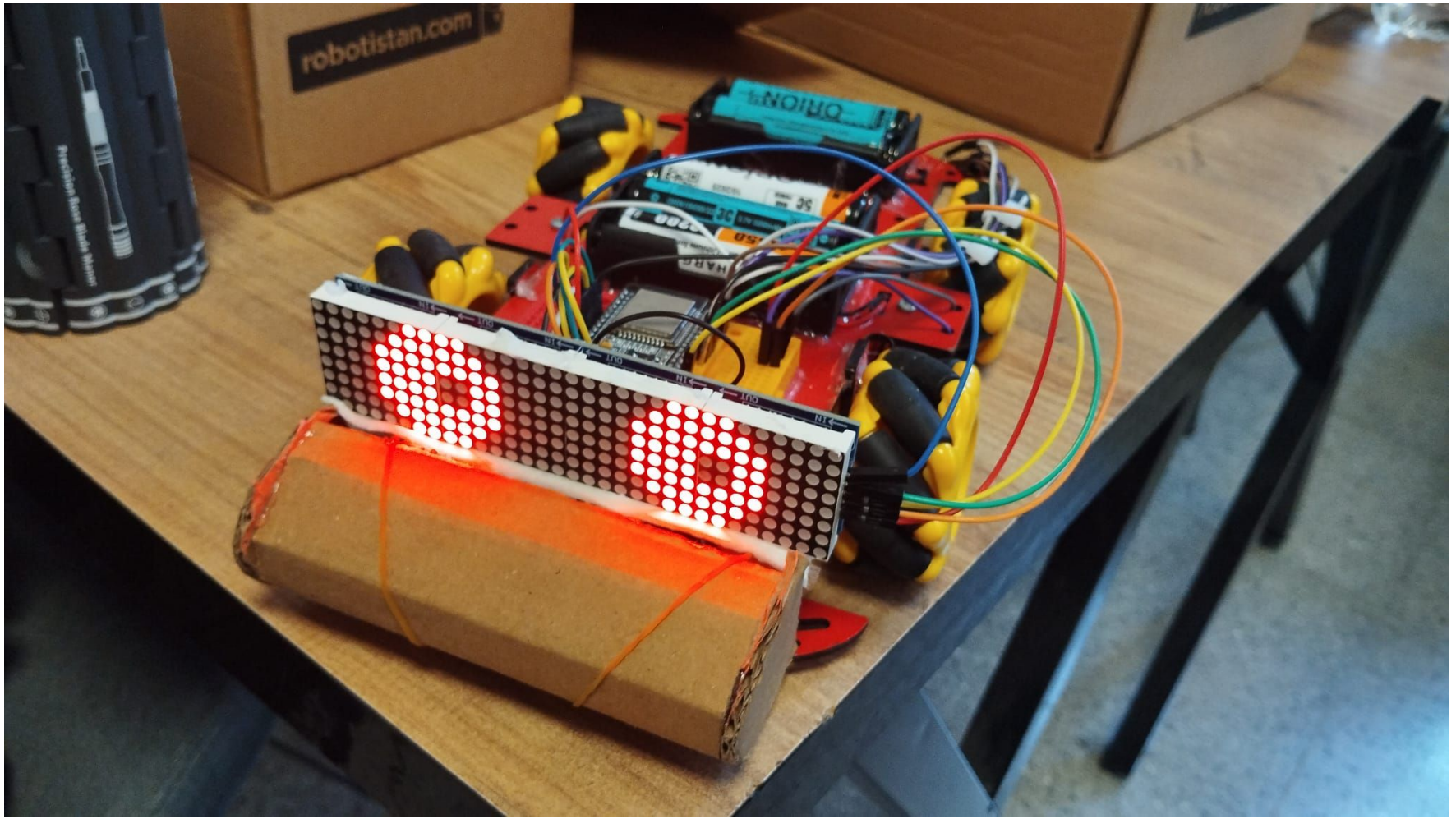
Stage 7 - Field Testing and Refinement

Operational testing will be performed in open environments to evaluate wireless range, latency, and physical responsiveness. Parameters will be fine-tuned based on the results.

Stage 8 - Evaluation, Documentation, and Final Report

Final system performance will be assessed and will be compared to the project goals. Technical documentation and codebase will be finalized, robot will be prepared for the final demonstration.

Evaluation
Final Product



Business Opportunities

Review

- ROBO REMIX

Pre-programmed dances

Button-based music games

LED face + sound effects

No user-controlled driving

Gesture-Controlled Robot Car

Hand-gesture driving

Basic forward/back/left/right movement

Usually no music system

Usually no facial/display feedback

- PartyBot

Real-time glove control

Mecanum wheel strafing

Music playback modes

Dot matrix expressions

More interactive entertainment focus

Business Opportunities
Comparison

- We proposed this project to enhance effectiveness of the tools in the entertainment sector with advanced controls more natural to the human movement. By achieving this natural movement, robot can be controlled without less muscle effort.

Conclusion

In the end we successfully constructed the PartyBot and passed all of the functional verification tests we prepared.
Overall functionality robot was on the expected range but performance of the components might vary in long term usage which affects functionality.

Demonstration

https://drive.google.com/file/d/1YzLhKe2rUd3AarrYWY_RYc2Ru71VwqNj/view?usp=sharing

